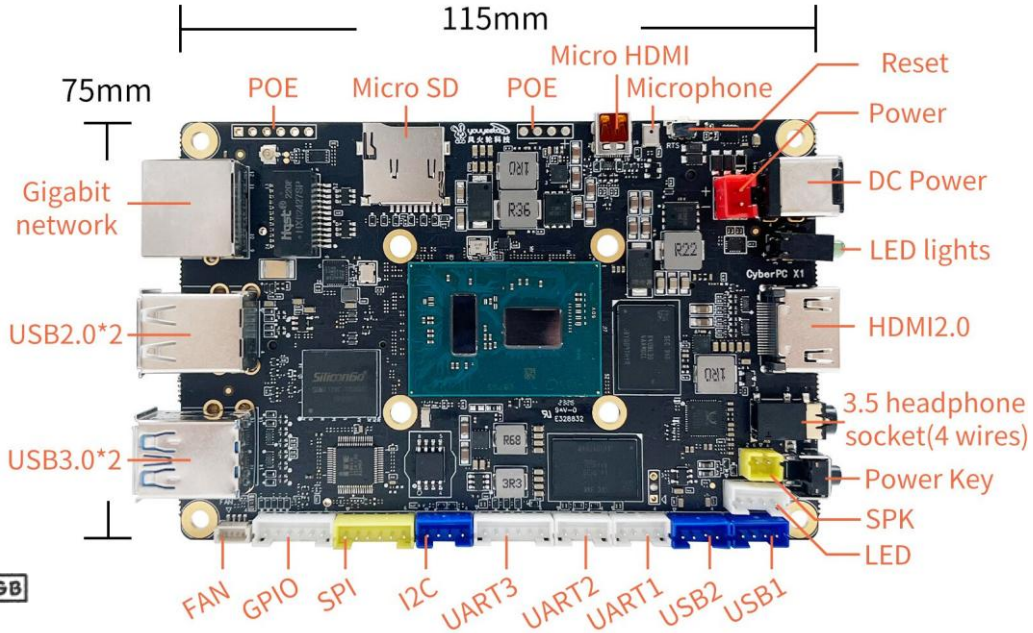
Embedded System

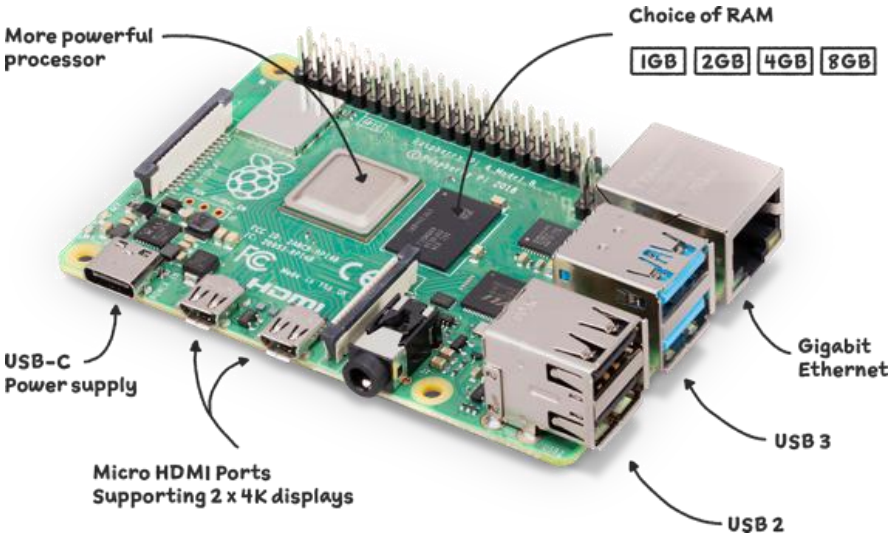
Ngô Lam Trung
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E-mail: trungnl@soict.hust.edu.vn

Chapter 2: Embedded Hardware

- What are the “things” on these boards?









Chapter 2. Embedded Hardware

- 2.1 Electronic parts and design tools
- 2.2 Organization of embedded systems
- 2.3 Sample development kits

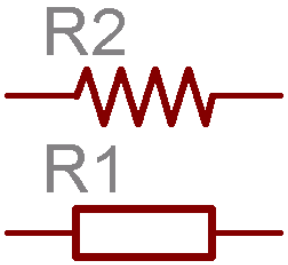
2.1. Electronic parts and design tools

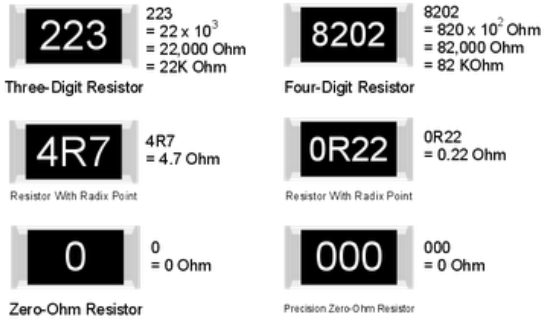
- 2.1.1 Basic parts
- 2.1.2 Integrated circuits
- 2.1.3 Tools

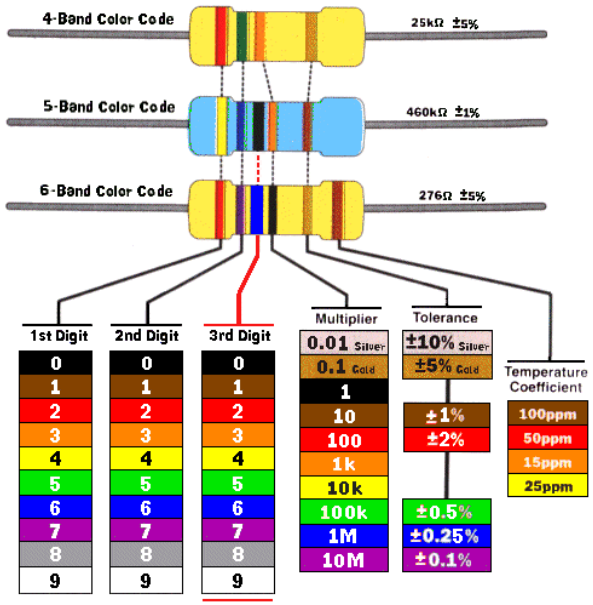
2.1.1 Basic parts

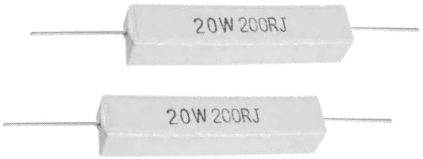
- Resistors
- Capacitors
- Inductors
- Diodes
- Crystal
- …

Resistors


R2
R1


223
223
= 22 x 10^3
= 22,000 Ohm
= 22K Ohm
Three-Digit Resistor
8202
8202
= 820 x 10^2 Ohm
= 82,000 Ohm
= 82 KOhm
Four-Digit Resistor
4R7
4R7
= 4.7 Ohm
Resistor With Radix Point
0R22
0R22
= 0.22 Ohm
Resistor With Radix Point
0
0
= 0 Ohm
Zero-Ohm Resistor
000
000
= 0 Ohm
Precision Zero-Ohm Resistor


4-Band Color Code
25kΩ ±5%
5-Band Color Code
460kΩ ±1%
6-Band Color Code
276Ω ±5%
1st Digit
2nd Digit
3rd Digit
Multiplier
Tolerance
Temperature Coefficient
0.01 Silver
0.1 Gold
1
10
100
1k
10k
100k
1M
10M
±10% Silver
±5% Gold
±1%
±2%
±0.5%
±0.25%
±0.1%
100ppm
50ppm
15ppm
25ppm

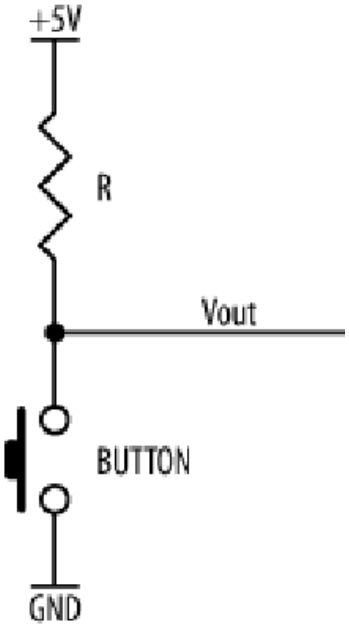

20W 200RJ
20W 200RJ


100W 2RJ

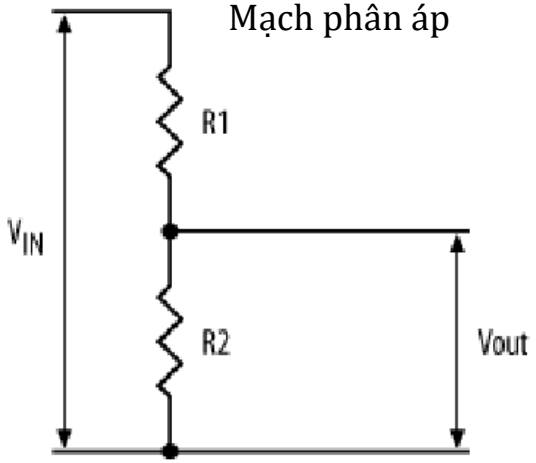

100W 2RJ

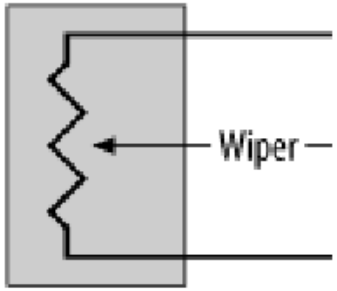
Resistors

- Sample usage:



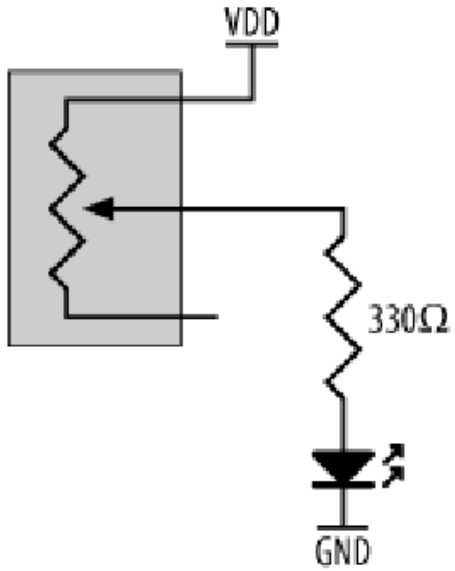
Pull-up resistor





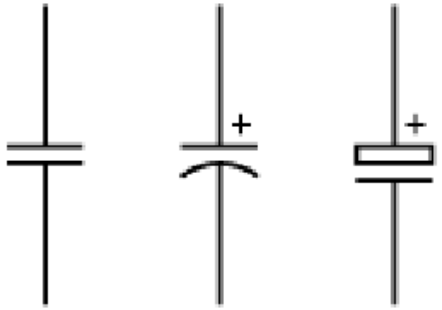


Biến trở
(potentiometer)

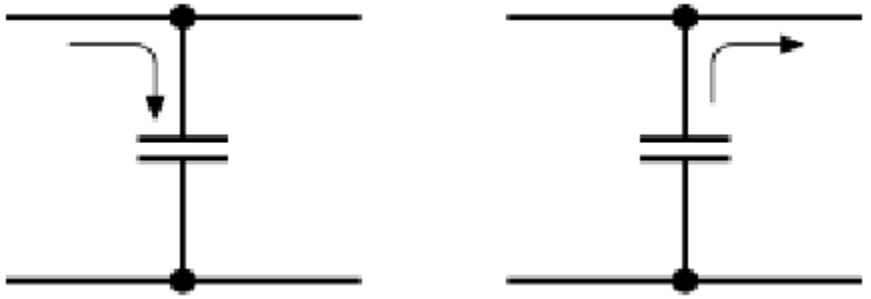


Capacitors

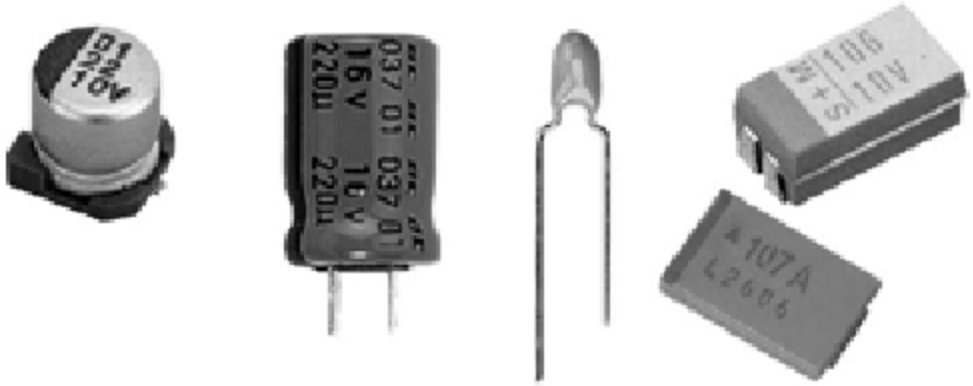
Symbol in schematics



Charging and discharging



Real capacitors

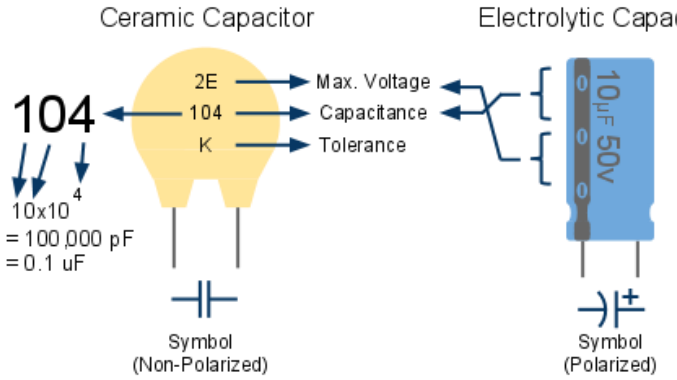




Capacitors

- Reading capacitor values

Capacitors



Capacitance Conversion Values

Microfarads (µF)	Nanofarads (nF)	Picofarads (pF)
0.000001 µF	0.001 nF	1 pF
0.00001 µF	0.01 nF	10 pF
0.0001 µF	0.1 nF	100 pF
0.001 µF	1 nF	1,000 pF
0.01 µF	10 nf	10,000 pF
0.1 µF	100 nF	100,000 pF
1 µF	1,000 nF	1,000,000 pF
10 µF	10,000 nF	10,000,000 pF
100 µF	100,000 nF	100,000,000 pF

Max. Operating Voltage

Code	Max. Voltage
1H	50V
2A	100V
2T	150V
2D	200V
2E	250V
2G	400V
2J	630V

Tolerance

Code	Percentage
B	± 0.1 pF
C	±0.25 pF
D	±0.5 pF
F	±1%
G	±2%
H	±3%
J	±5%
K	±10%
M	±20%
Z	+80%, -20%

Inductors

Schematic symbol

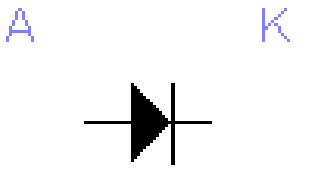




Real inductors

Diodes

- Diode:
  - symbol:



Power diode (1N4004)

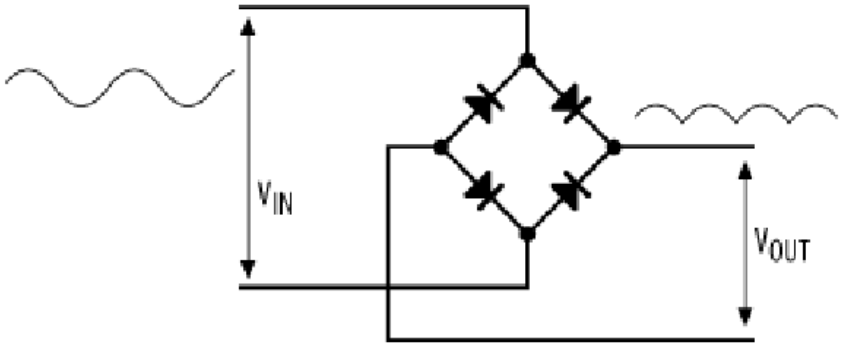


  - Conducts current in one direction: A to K
  - Hoạt động:
    - Nếu $U_A > U_K$ thì $I_{AK} > 0$, Diode làm việc ở chế độ Thông
    - Nếu $U_A \leq U_K$ thì $I_{AK} = 0$, Diode làm việc ở chế độ Tắt

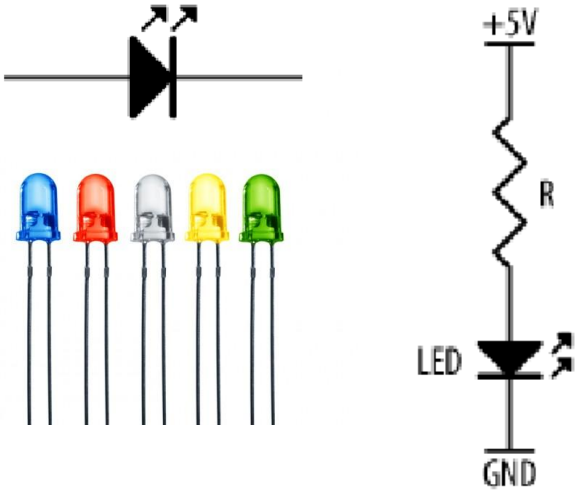
Diodes

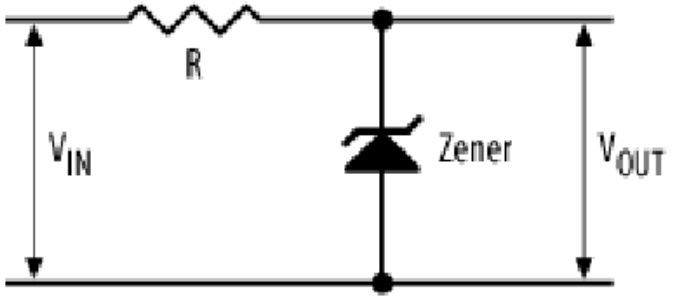
- Sample usage

Bridge rectifier

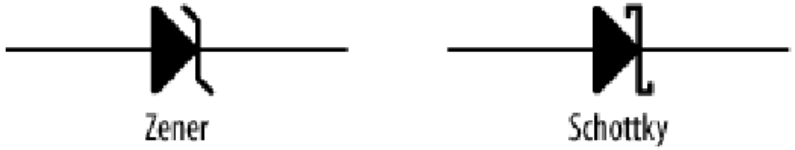


LED





Zener and Schotky diodes



Crystal

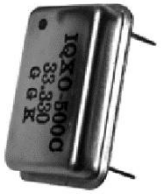
- Generate clock signal with stable frequency

Ký hiệu

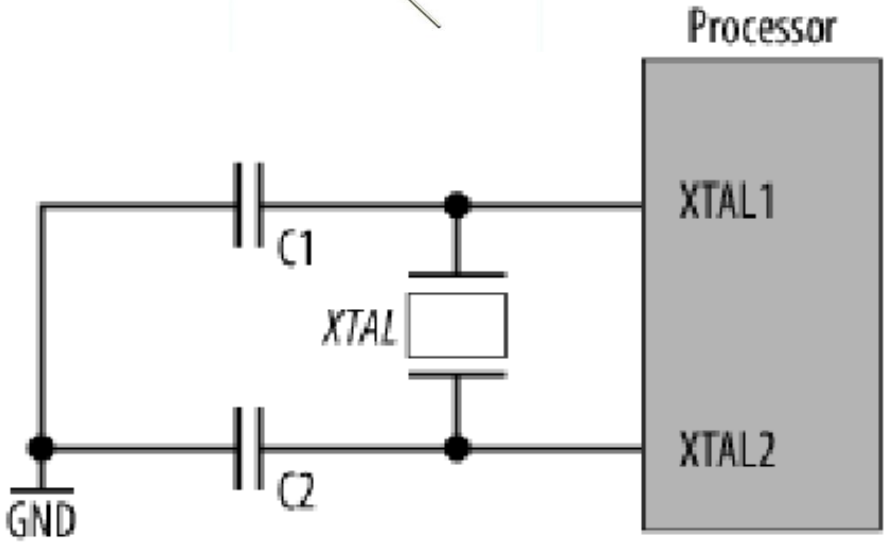












2.1.2 Integrated circuit (IC)

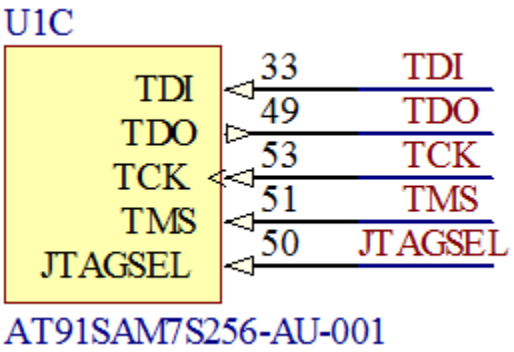
- Microprocessor, Microcontroller
- Power regulator
- Memory
- Sensors
- Communication
- Other functional ICs
- …

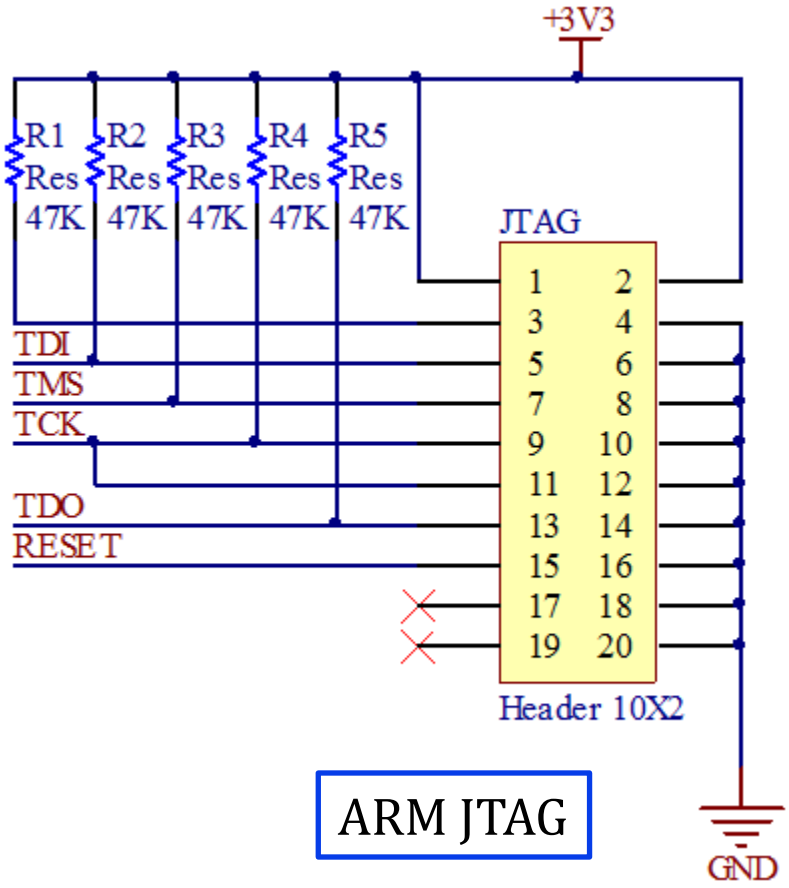
2.1.3 Tools and equipments

- Programmer and debugger
- Multimeter
- Osciloscope
- Power source
- Tools: solder station, nipper, wire striper,...

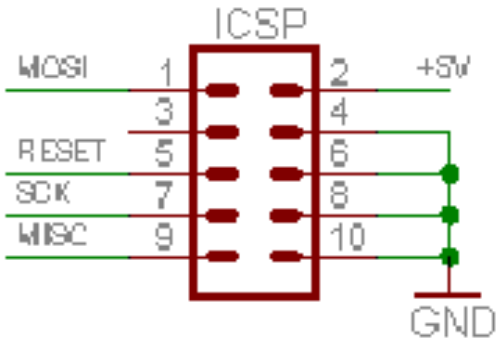
2.1.3. Programmers/Debuggers

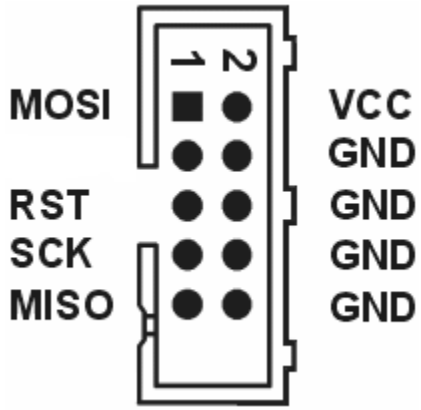
- "Burn" the firmware to IC using JTAG, SWD, ISP, …
- Must be compatible with the MCU as a part of the tool suite.





AVRISP





ARM JTAG

Multimeter

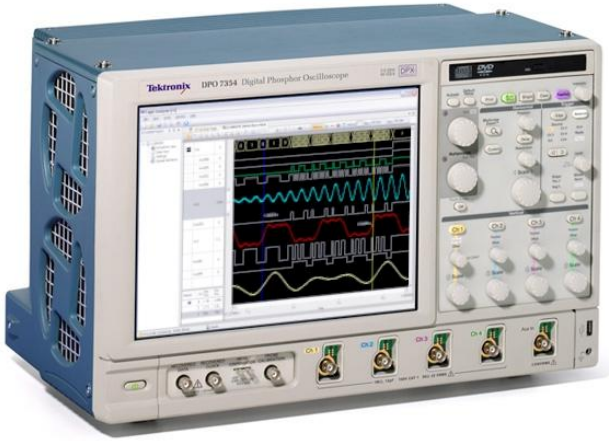
- Handy tool to measure values of voltage, current, resistor, capacitor,…

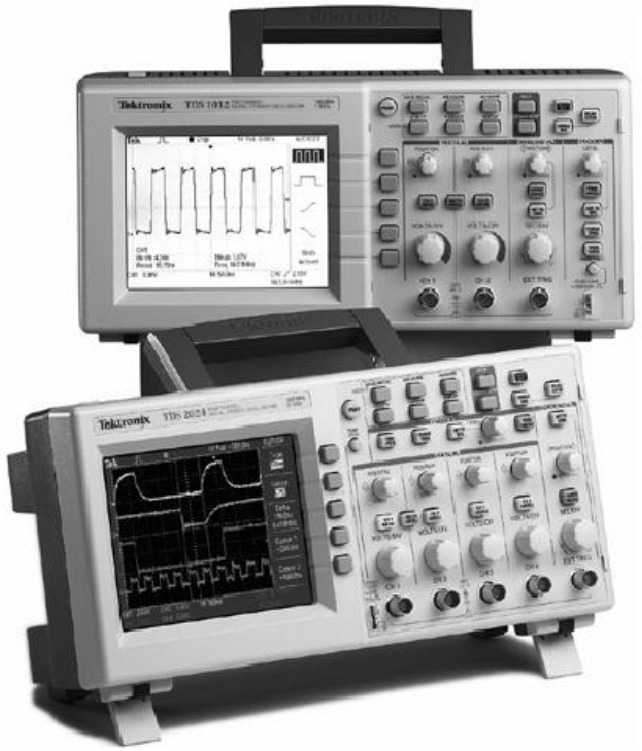




Oscilloscope

- Measure and display the waveforms of signals
- Example: Tektronik TDS5104
  - 1 GHz bandwidth, 4 channels
  - 5 GSs (giga samples/s)





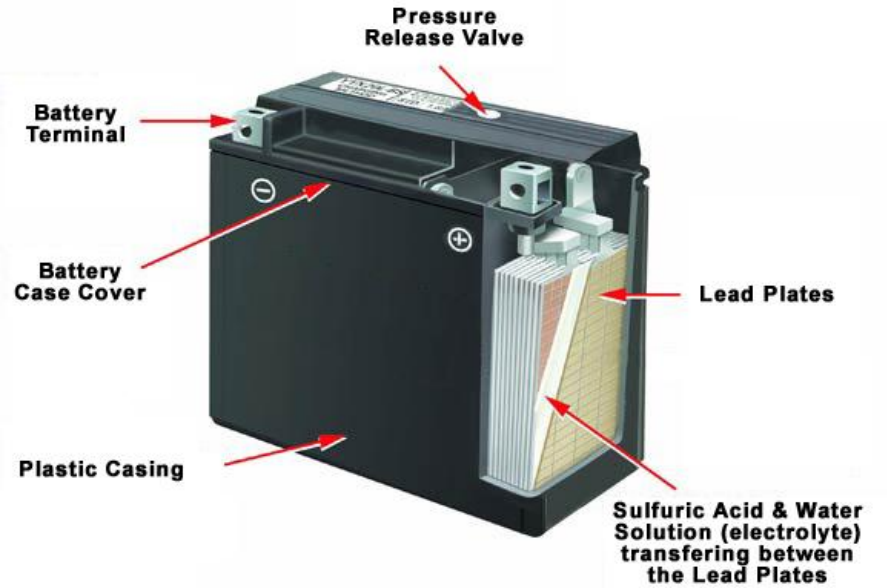
How to use an oscilloscope ? Video: (Tektronik tutorial)
http://www.youtube.com/watch?v=tzndcBJu-Ns&list=PLF6BB1B46F7AD8751

Power supply unit

- Generate regulated DC power



Batteries & Acquy



AC-DC Adapter
Input: AC 100-240V
Output: DC 5V/6V/9V, …
500mA/1A/2A, …

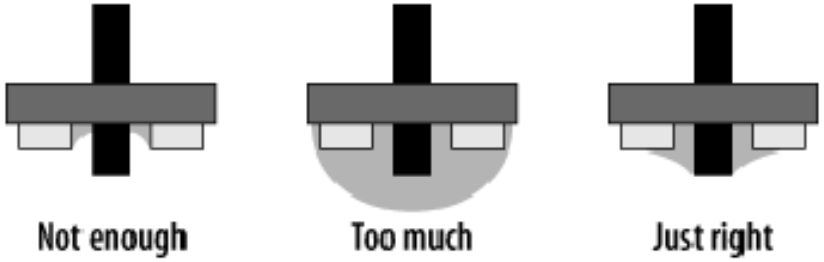




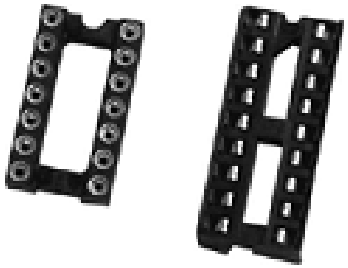
Alkaline
Zinc-carbon
Lithium (Li-ion)
Nickel-cadmium (NiCd)
Nickel metal hydride (NiMH)
Lead-based

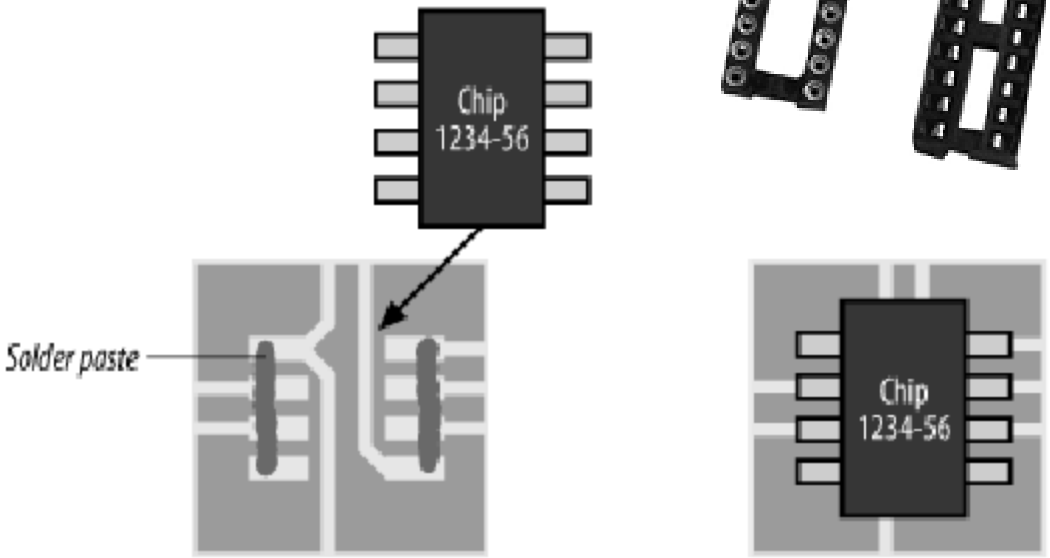
Solder

- How to solder ?







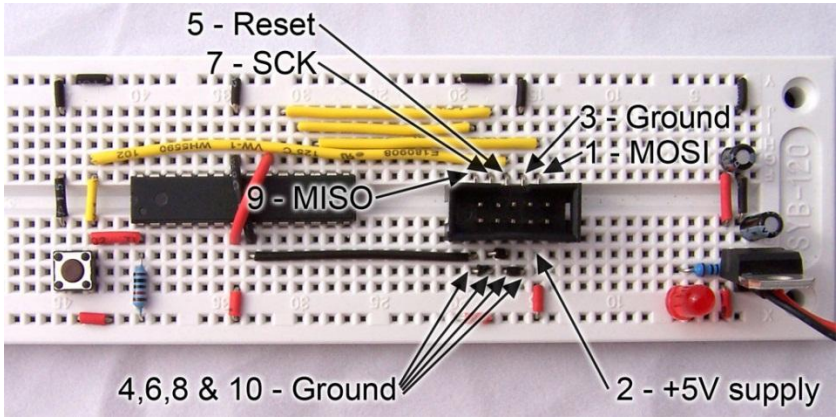


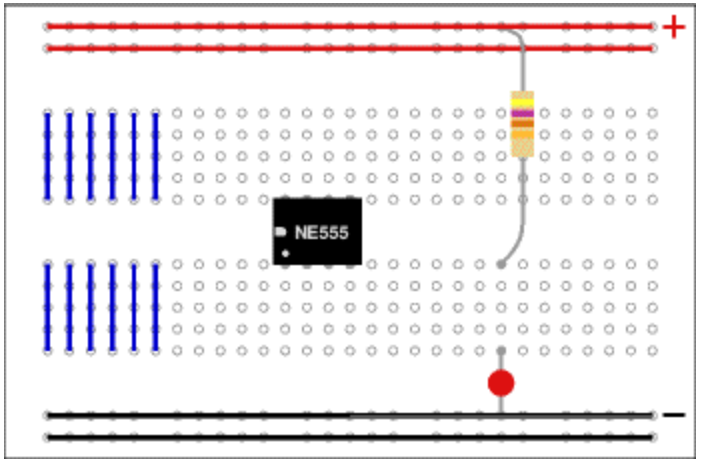


Other tools



Breadboard

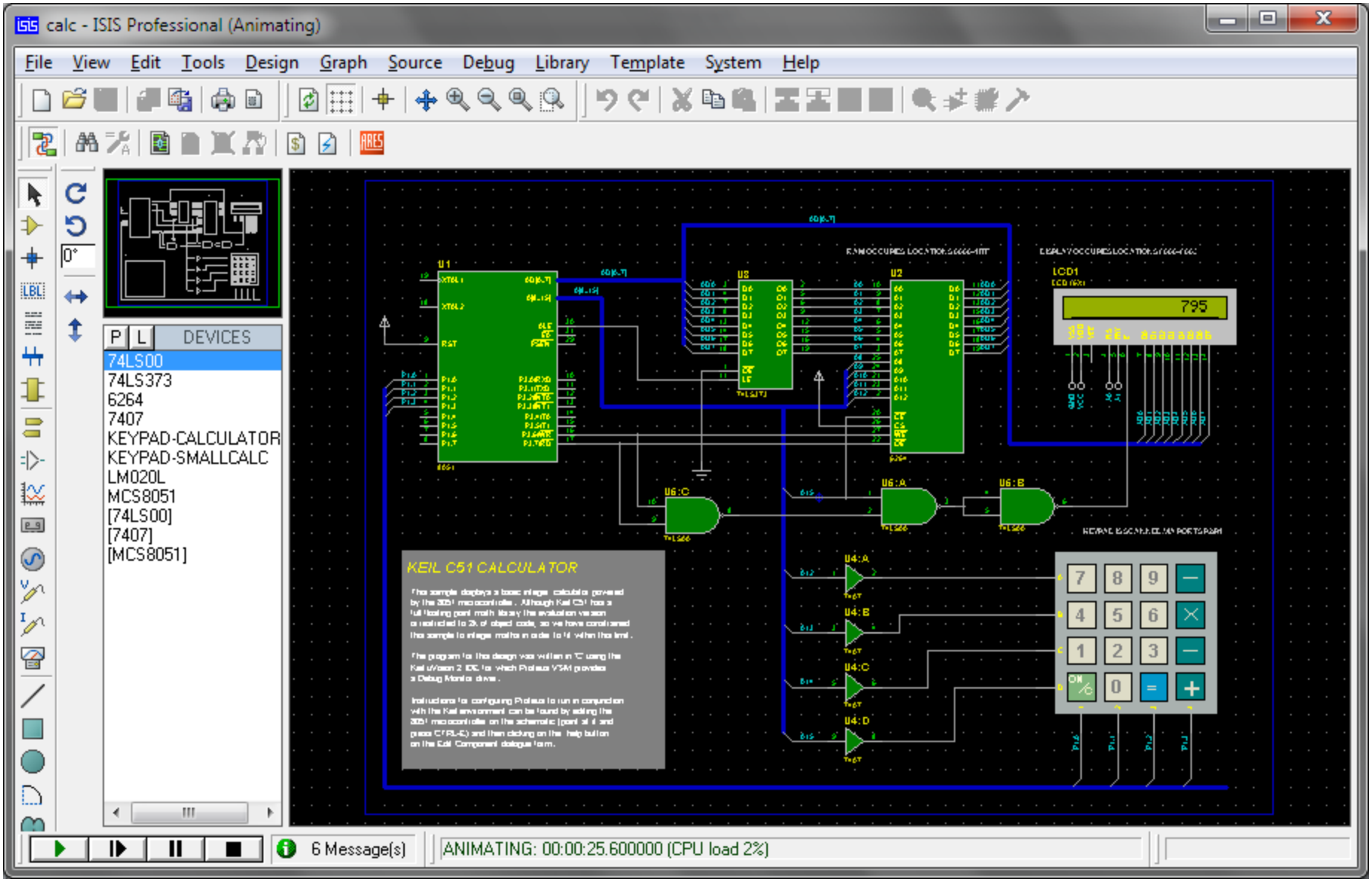




IDE and simulator

- Offine:
  - Proteus, Multisim,…
  - Arduino, Platform IO,…
  - STM32CubeIDE, Code Composer Studio,…
- Online: wokwi, TinkerCad, EasyEDA,…
- Features :
  - Design schematics and simulate operation
  - Support component library: microcontrollers, logic gates, resistor, capacitor, button, led, 7 seg, LCD, v.v…
  - Develop and debug system firmware

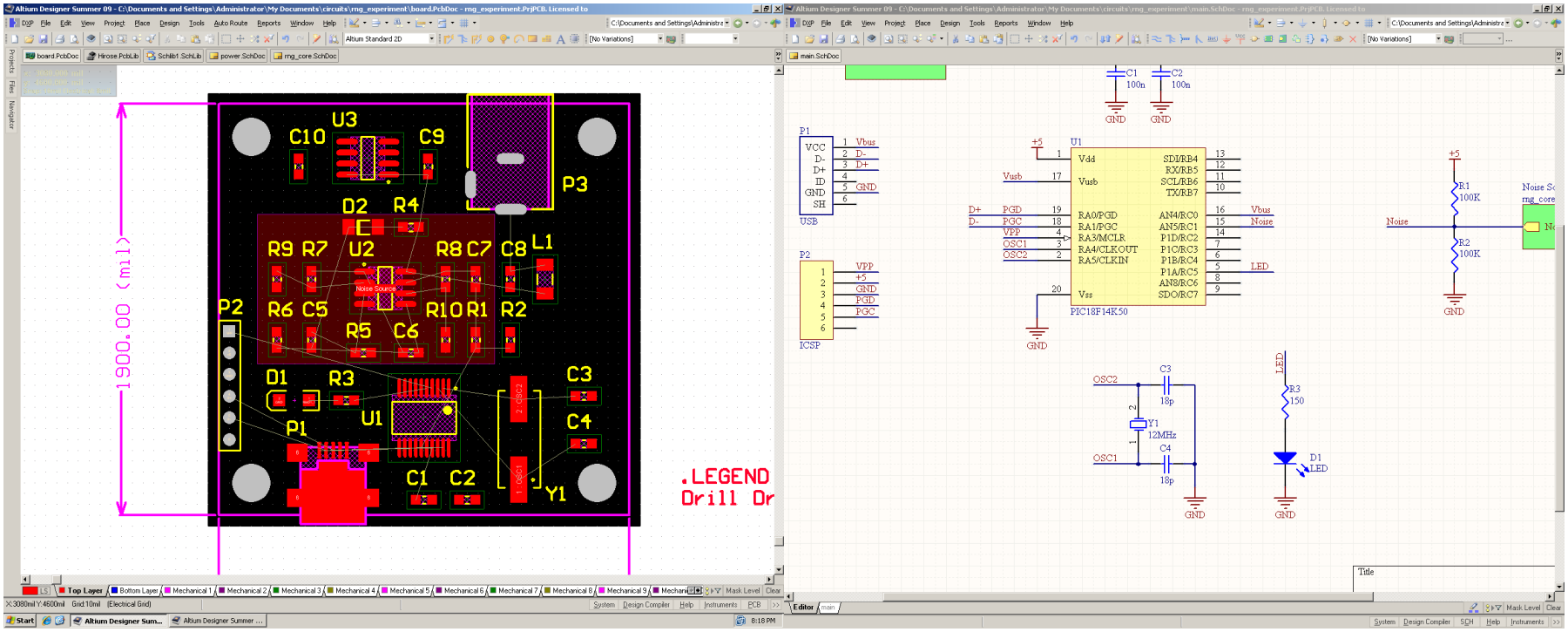
Example: Proteus



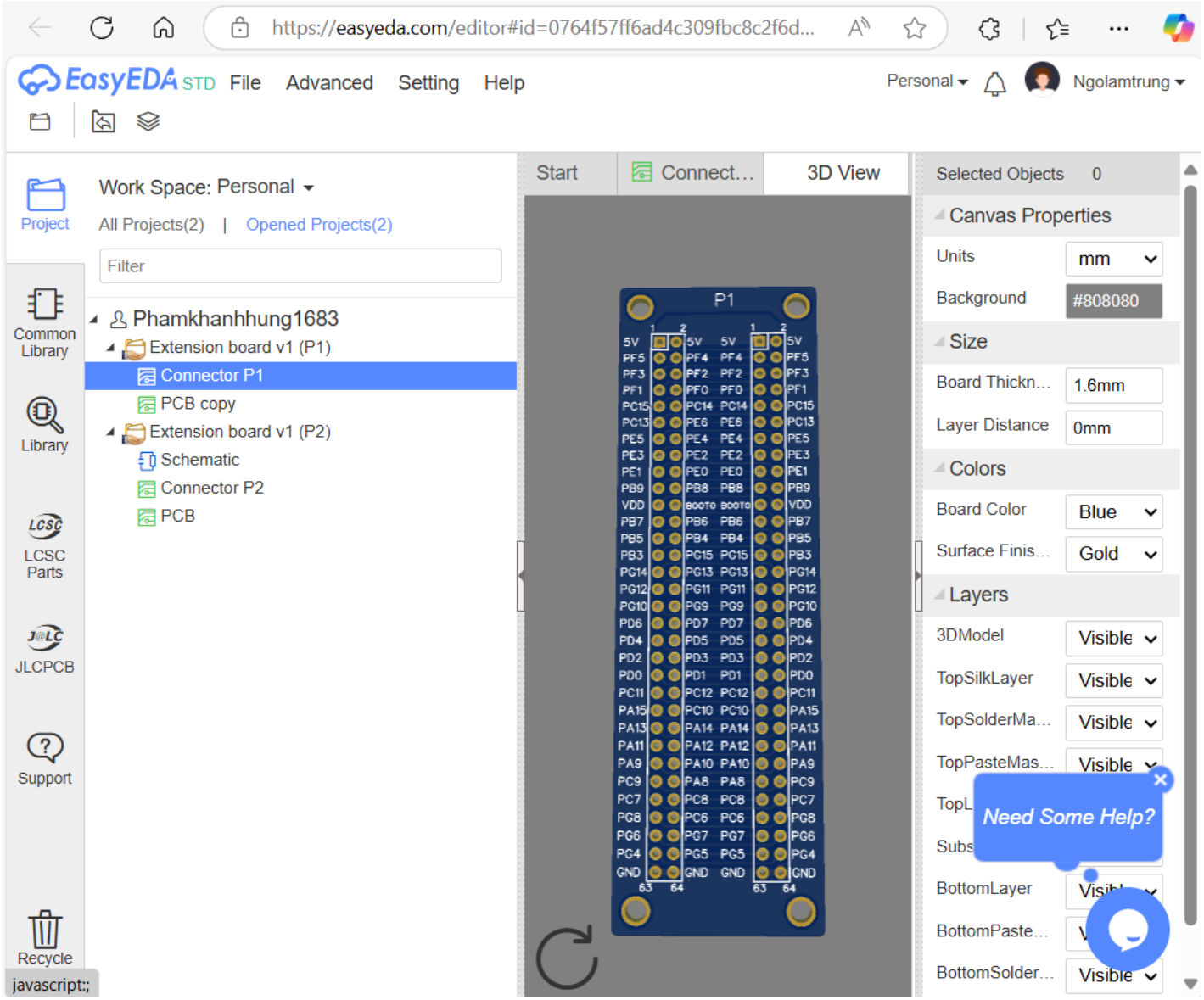
Example: Altium/Protel

- Popular tool for
  - Designing schematics
  - Design PCB
  - With a huge component library





Example: EasyEDA.com



Exercise: a simple circuit

- Create account on easyeda.com
- Design a new simple circuit

2.2. Basic components of embedded systems

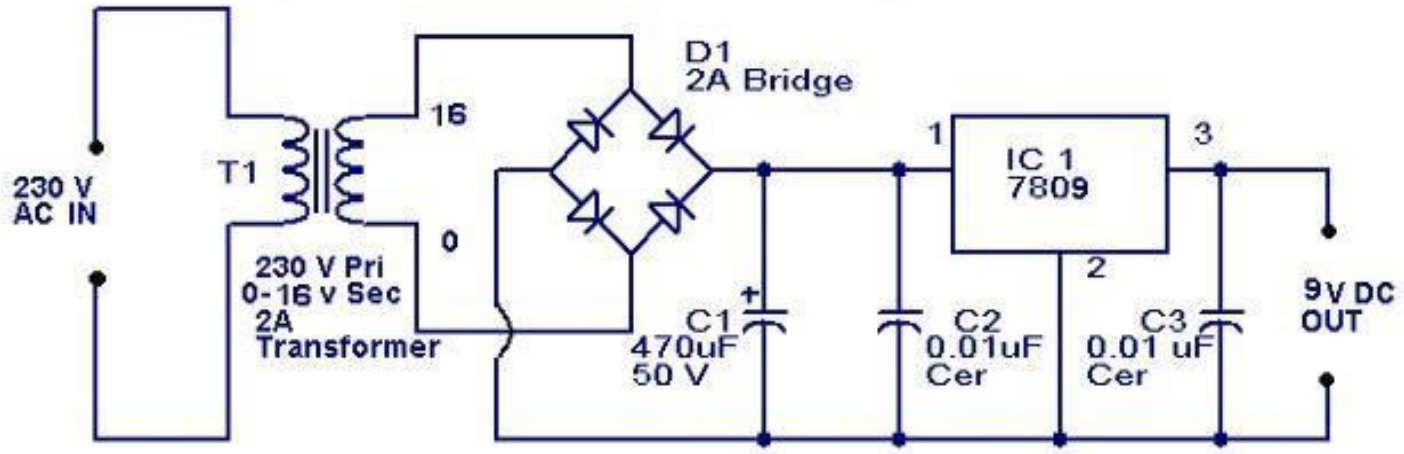
- 2.2.1 Central system
  - CPU and memory
  - Reset
  - Power source
  - Oscilator
- 2.2.2 Peripherals
  - Digital interface
  - Communication
  - Analog interface

2.2.1 Central system

- Computing module, consists of CPU/MCU and the most necessary components to make the CPU/MCU work and execute program.
- Available either as a separate module, or built together with the whole component on a single circuit.
  - Arduino: can be considered a separate central system.
- CPU/MCU: small system usually uses MCU with integrated memory and basic I/Os.
- Memory: integrated on the CPU/MCU IC, and/or with separate ROM/RAM.
- eg: chip STM32F429ZGT6
  - CPU ARM Cortex-M4 max 180 MHz
  - Built-in 1 MB ROM, 260 KB SRAM
  - External DRAM supported

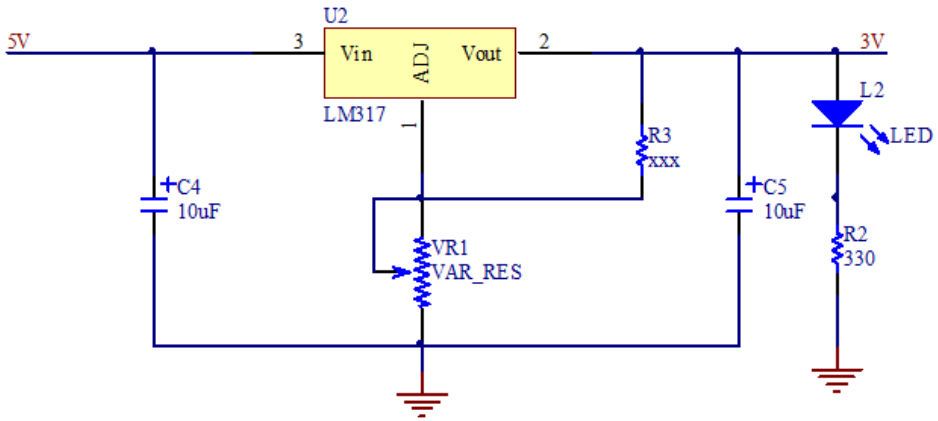
2.2.1. Power Source

- Provide necessary stable power source for the whole system
- Basic components
  - Transformer
  - Rectifier
  - Voltage regulator

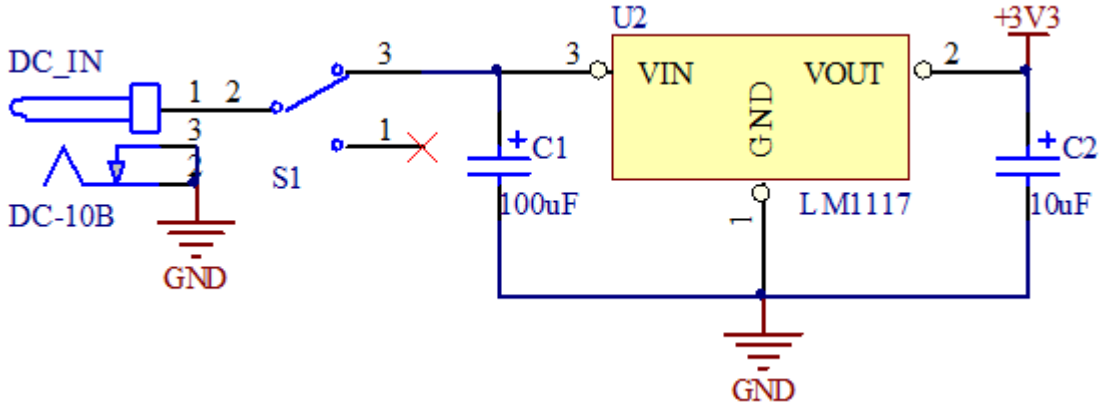


Power Source

- Voltage regulator IC.
  - Input: DC voltage that may fluctuate
  - Output: stable DC voltage
  - Fixed our adjustable V_{out}
- Some popular IC: 7805/7809, LM317/1117,

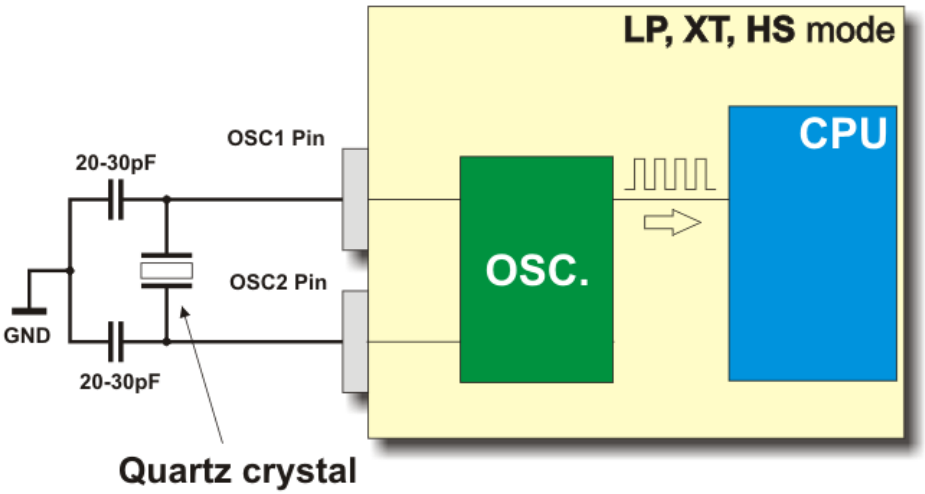


Vout = 1.25*(1+VR1/R3)



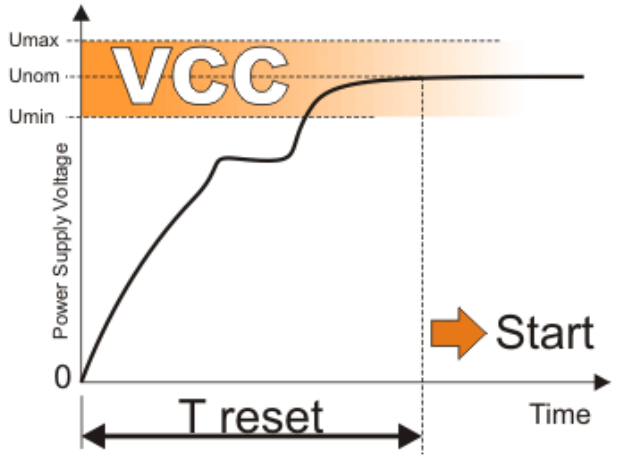
Oscilator

- Generate the square wave that is used as the clock signal of sequential circuits
  - CPU
  - Real-time clock, USB IC, Ethernet IC,…
- Internal oscilator: RC resonator built-in the MCU, convinient but less stable.
- External oscilator: exter crystal IC as resonator with much more stable clock rate.



Reset circuit

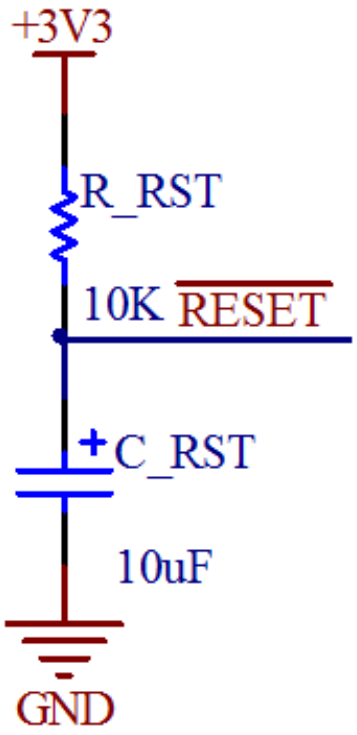
- Reset signal:
  - Mandatory to bring up all CPU/MCUs.
  - Generated by external reset circuit, or internal mechanism
- When in reset condition: CPU is initialized to safe initial status
  - PC: reset vector address
  - Other registers: clear to default values (see CPU specs.)
- Example: when the voltage of the reset pin stays at active level for at least the period of T_{reset}, the CPU will enter reset condition.
  - After that, the reset pin must return to inactive (for the CPU to work).

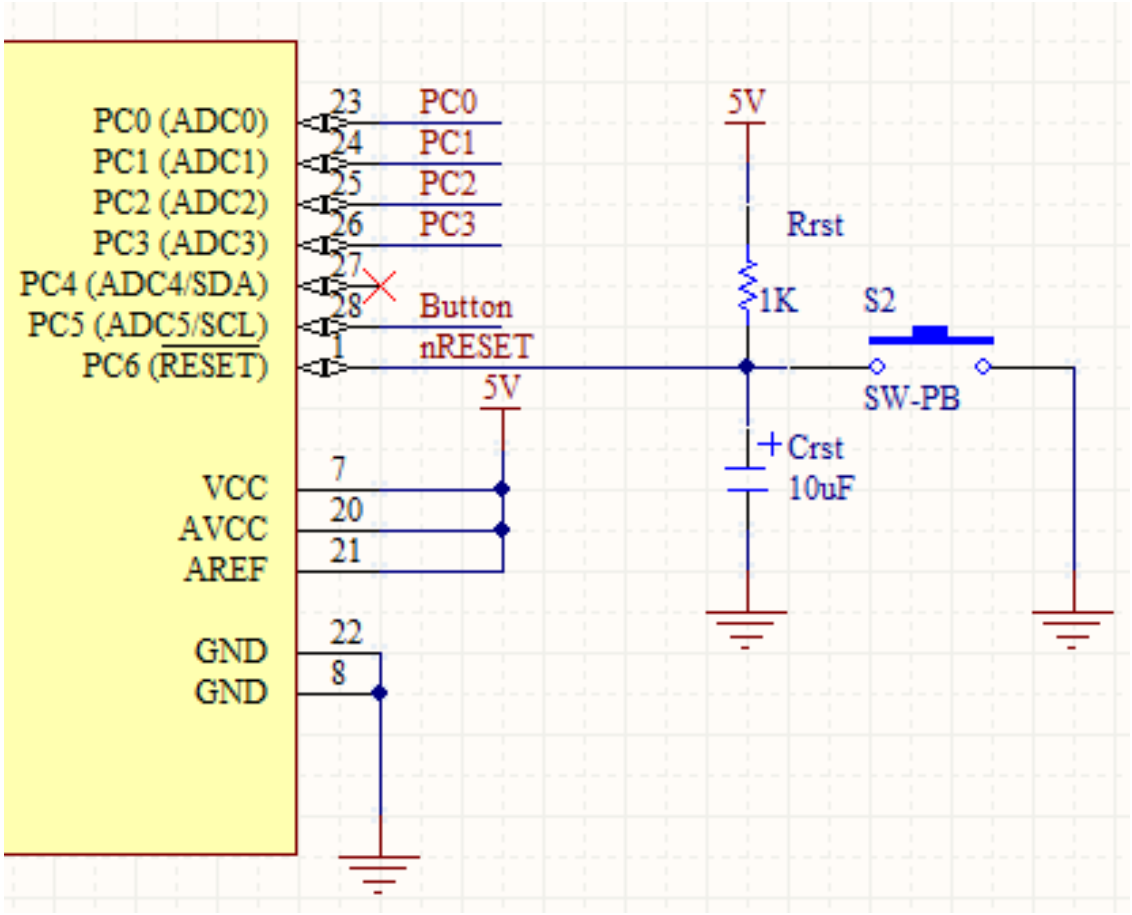


Reset when Power on

Reset

- Example: a reset circuit for with auto and manual reset



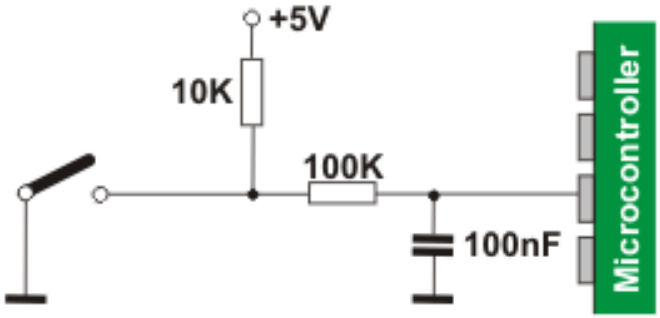


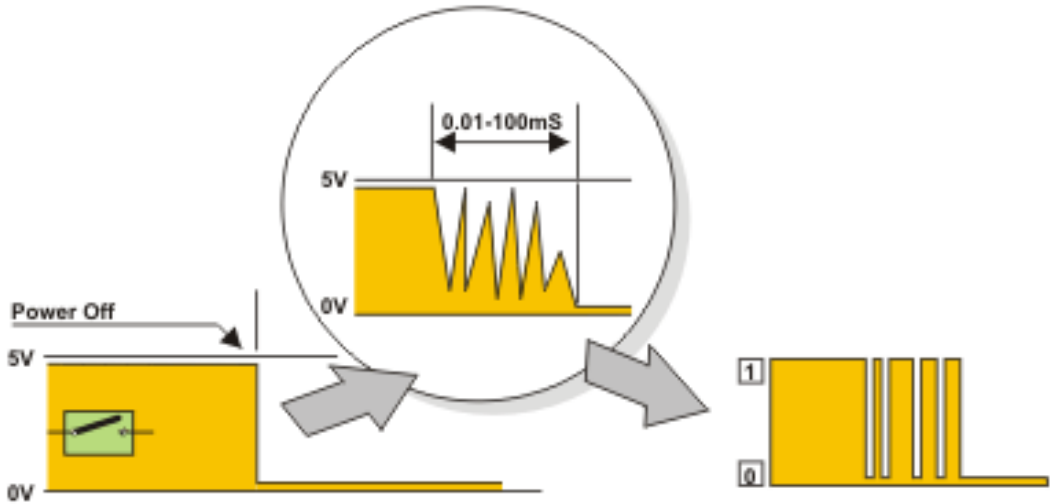
2.2.2 Peripherals and Interface

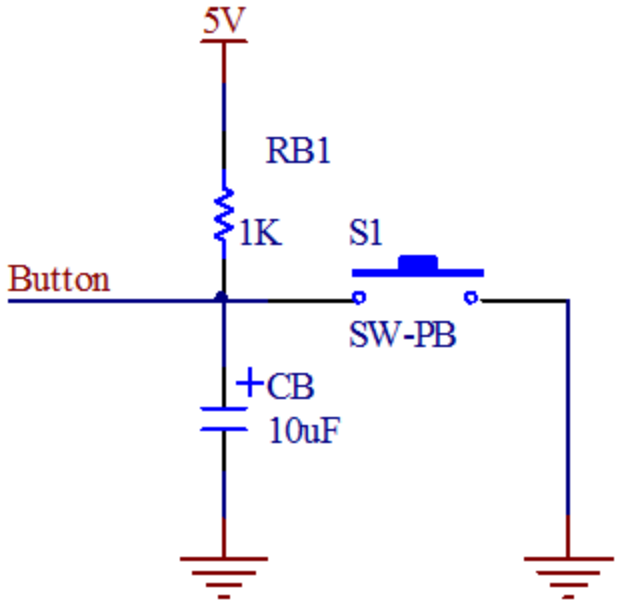
- Digital interface: switch, button, LED, LCD,…
- Communication: UART, SPI, I2C, USB,…
- Analog interface: DAC, ADC,…

Switch, Button

- Switch: switch on/off (permanently)
- Button: switch on/off (temporarily)
  - Key bouncing

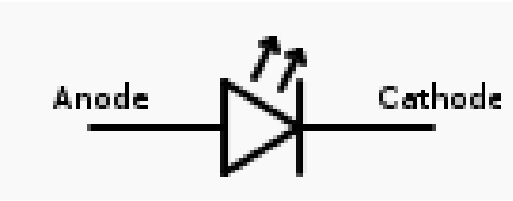


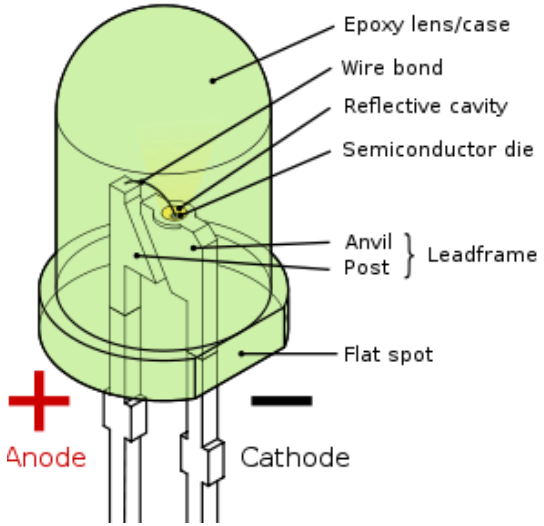


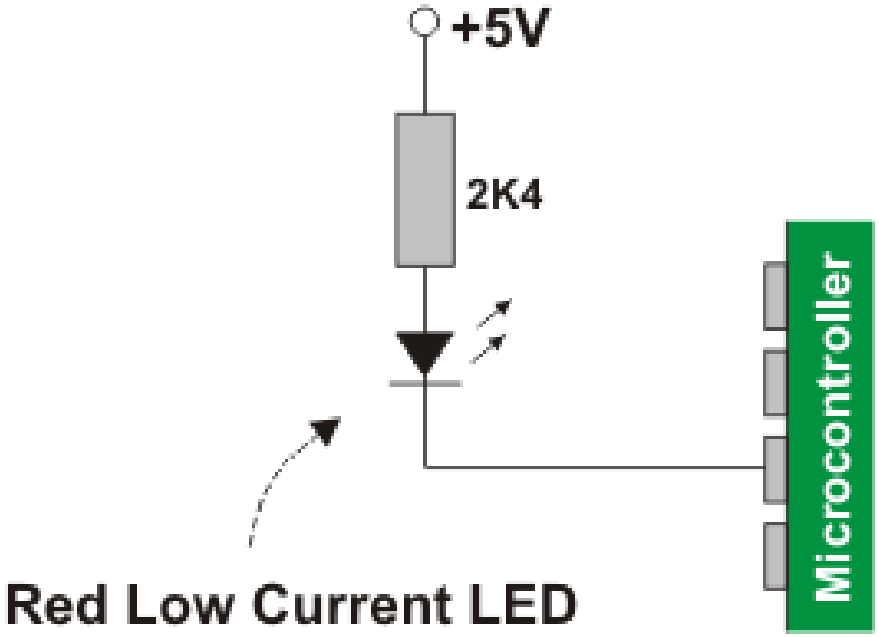


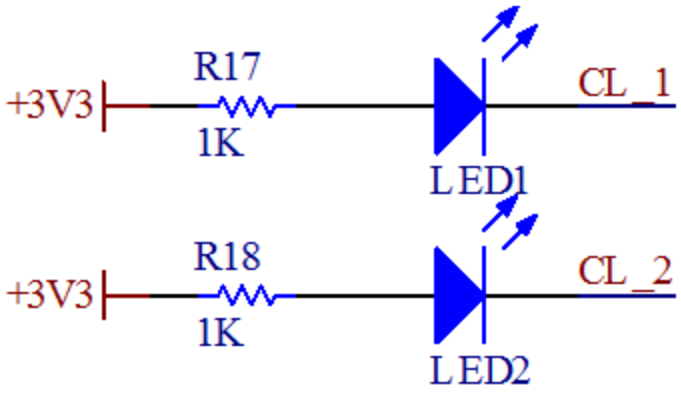
LEDs

- LED = Light-Emitting Diodes



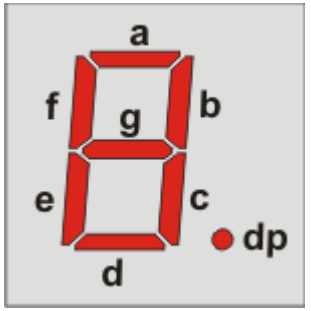


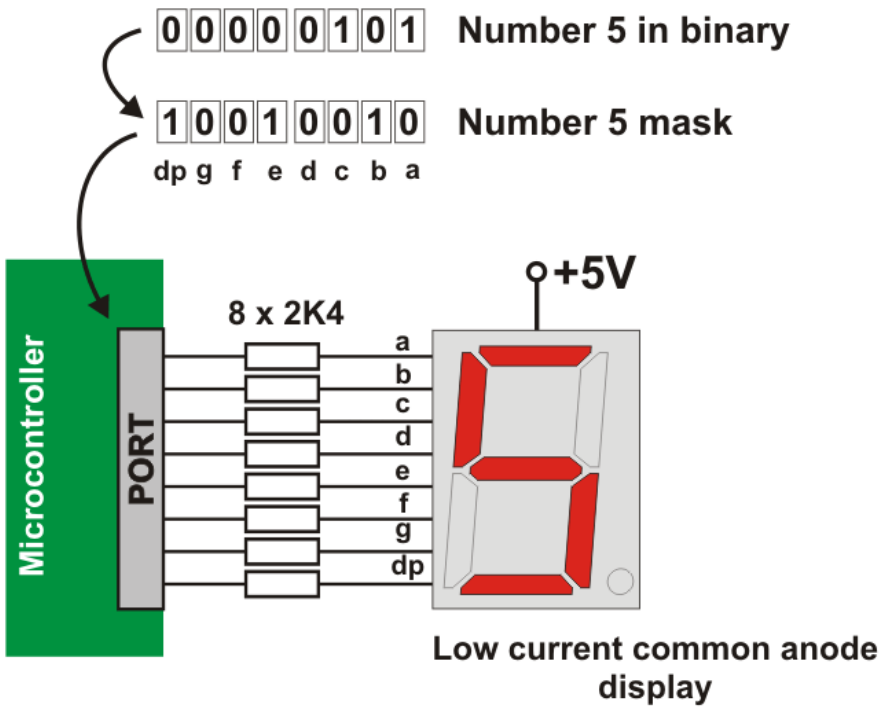




7-segment LED

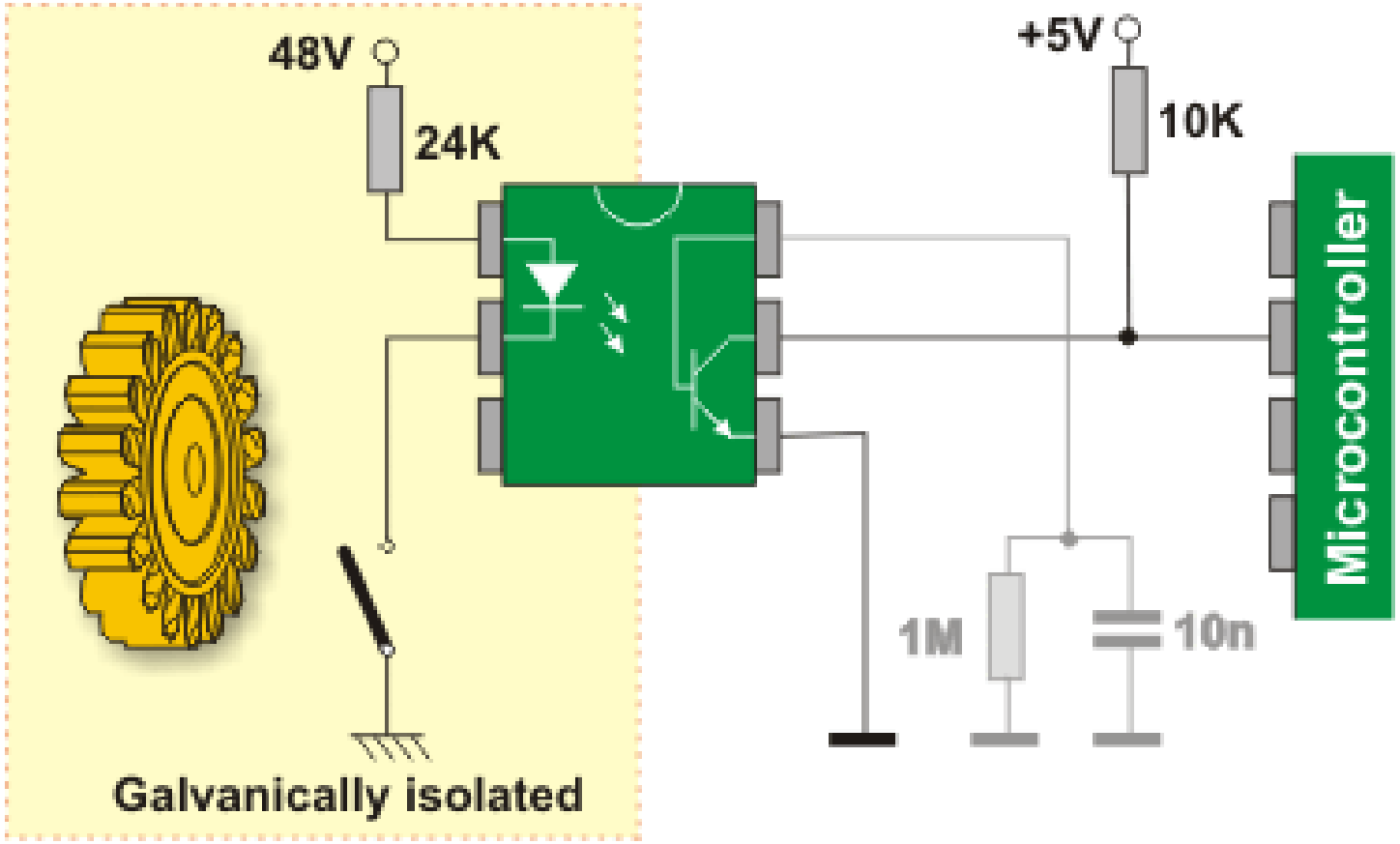
- 7-segment module: combination of 8 LEDs (a,b,c,d,e,f,g,dp)
- 2 types:
  - AC (Anode Common),
  - CC (Cathode Common)





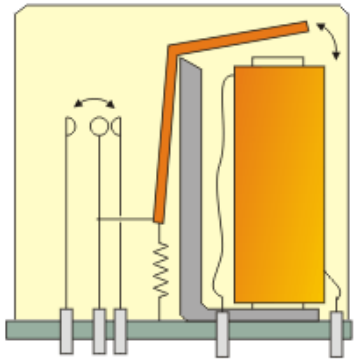
Opto-coupler

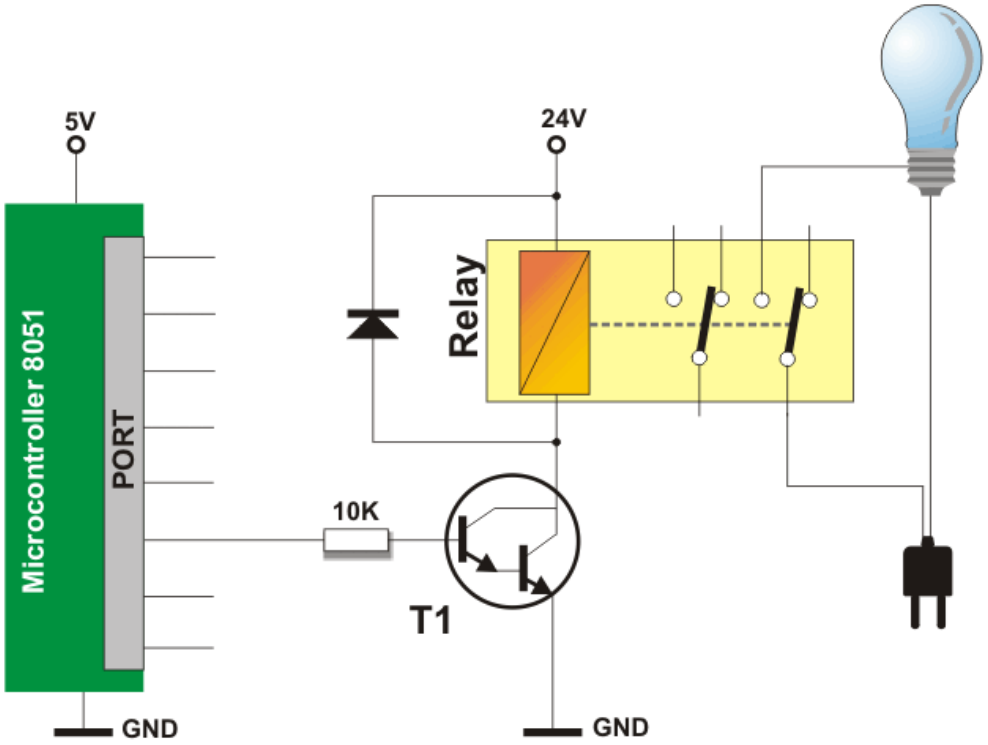
- Isolate the two circuits by using light.

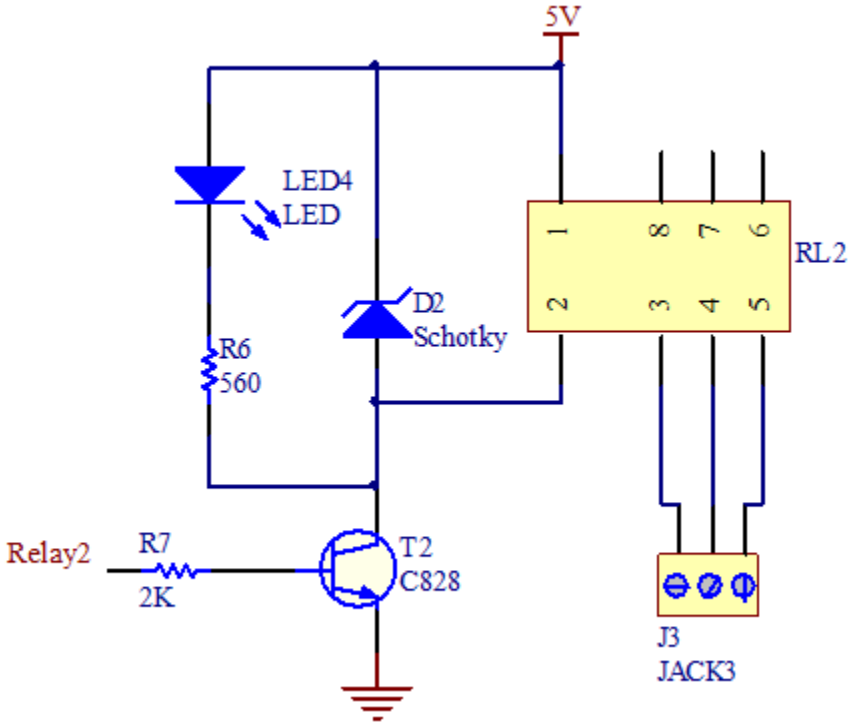


Relay for high power interfacing

How relay works





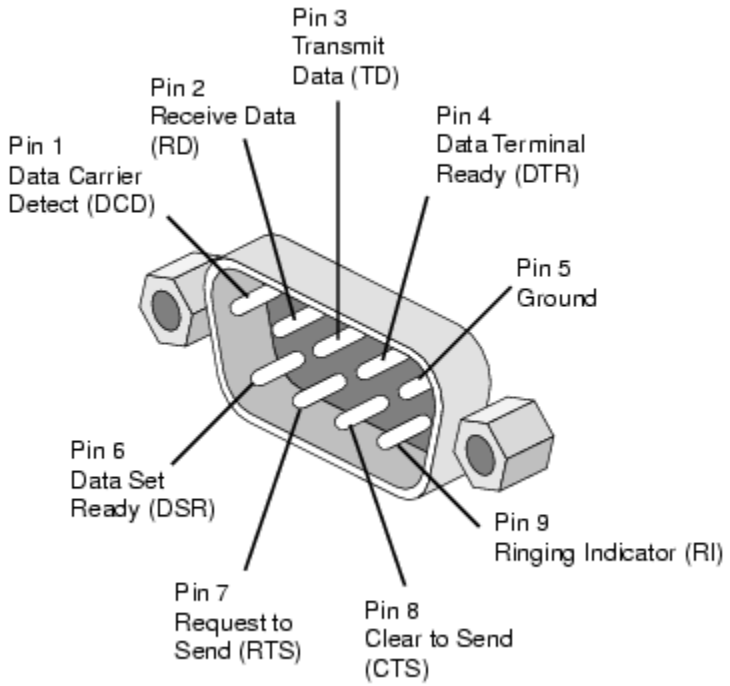


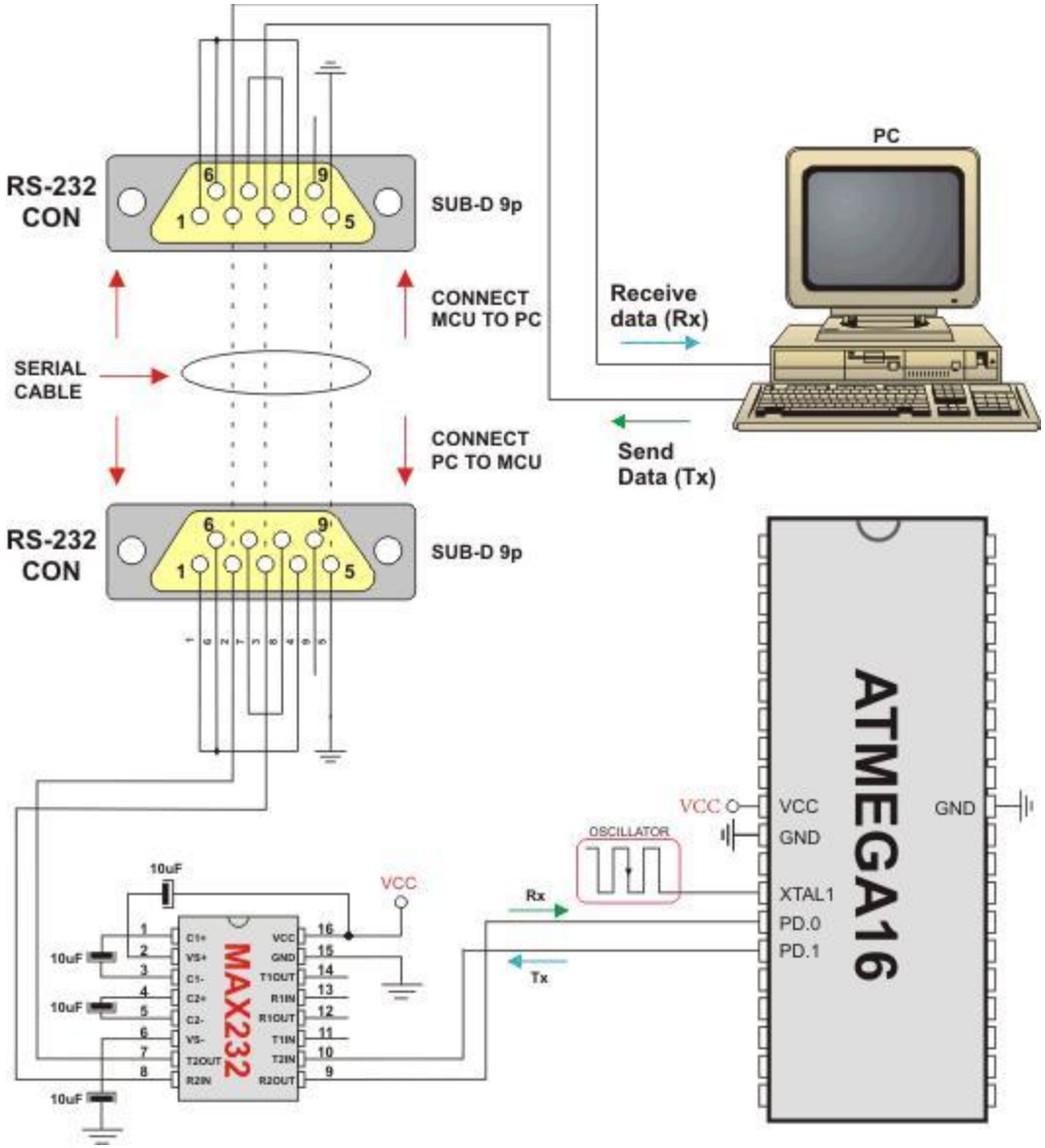
Communication

- RS232
- SPI
- I2C
- USB
- CAN…

RS232

- Serial Communication
- UART (Universal Asynchronous Receiver and Transmitter)
- Full Duplex

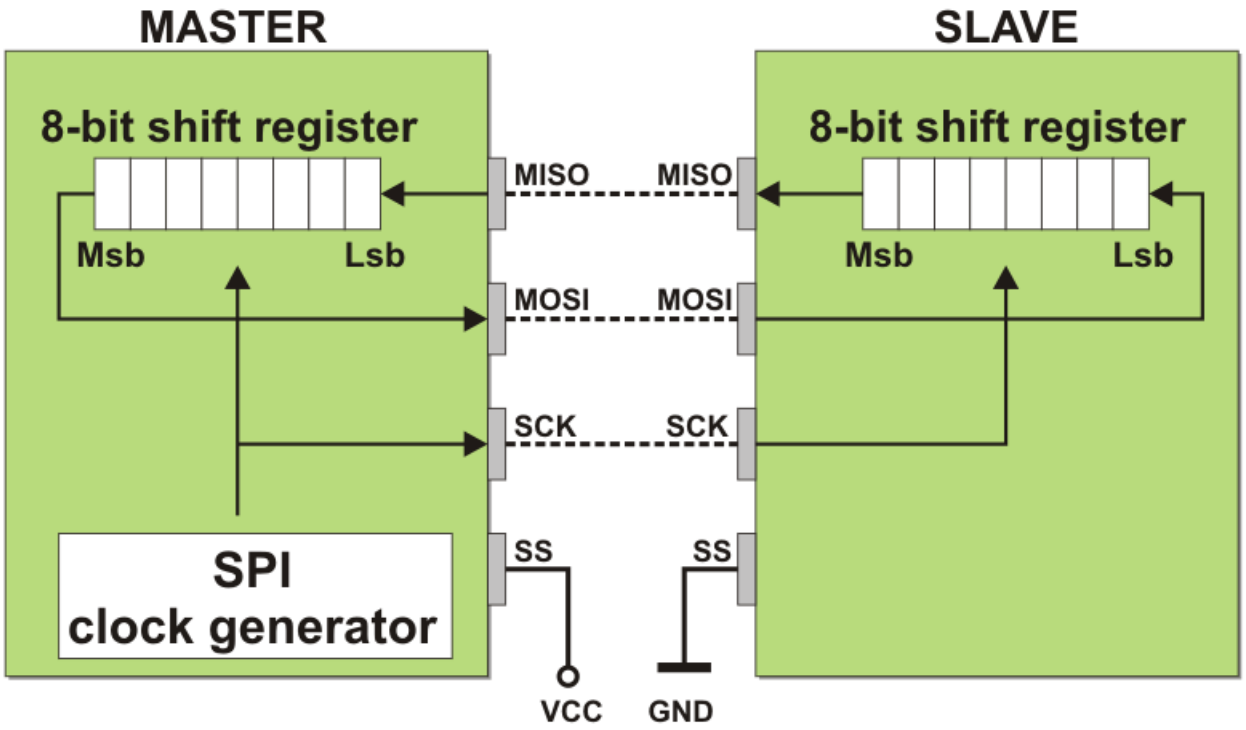




SPI

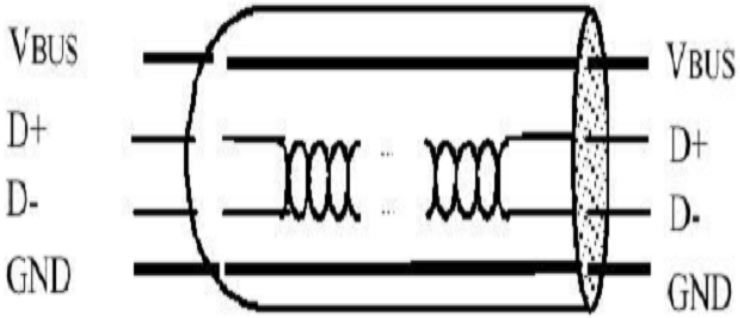
- SPI (Serial Peripheral Interface)
  - MCU – MCU interface
  - MCU – Devices (EEPROM, display,…)
- Master vs Slave

MOSI = Master Out Slave In
MISO = Master In Slave Out
SCK = Serial Clock
SS = Slave Select



USB

- USB=Universal Serial Bus
- Differential transmission





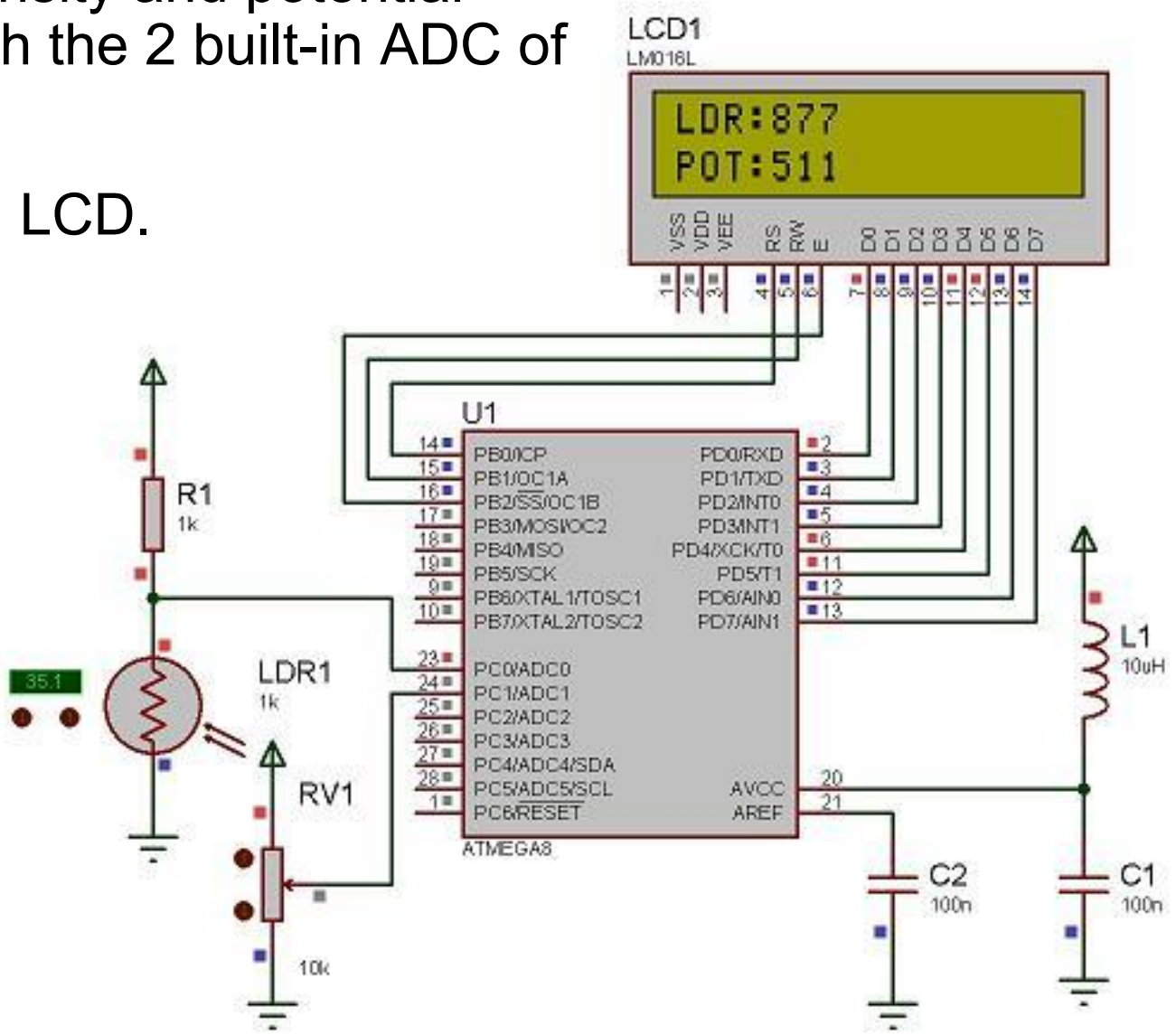
Pin	Signal	Desc.	Color
1	VBUS	+5 VDC	Red
2	D-	Data -	White
3	D+	Data +	Green
4	GND	Ground	Black

Analog interface

- Interface the embedded system with analog signal.
  - Measure analog signal: temperature, humidity, light intensity,...
  - Output analog signal: sound, light,…
- ADC: analog to digital converter.
- DAC: digital to analog converter.
- Selection of ADC/DAC:
  - On-chip: built-in as a peripheral on MCU.
  - Dedicated ADC/DAC chip: for higher performance

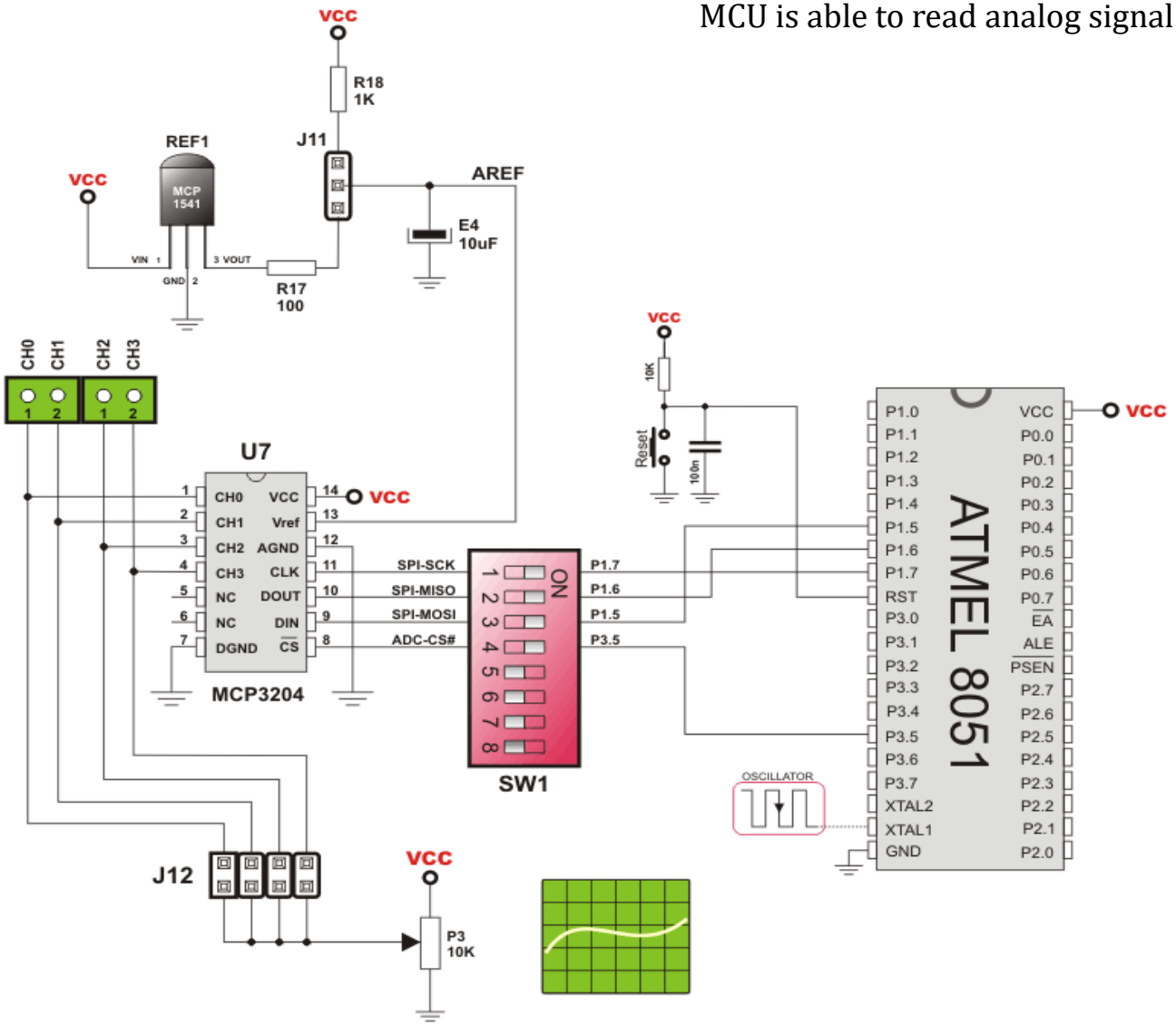
Example: built-in ADC

- Reading light intensity and potential meter position with the 2 built-in ADC of Atmega8.
- Display values on LCD.



Example: interfacing a 8051 MCU with MCP3204 ADC

MCU is able to read analog signal from 4 channels



2.3 STM32F429I-DISC1 development board

- STM32F429ZIT6 ARM Cortex-M4, 180 MHz max.
- 2 MB flash, 256 KB SRAM
- 8MB SDRAM
- On-board ST-Link debugger
- 2.4” QVGA touch LCD
- 6 LEDs (2 user LEDs)
- 3 axis gyros



Homework

- Read the STM32F429 datasheet
- Set up tool suite for STM32F4 firmware development
  - STM32CubeIDE 2.2
  - STM32CubeMX 6.18.1 + MCU Package for STM32F4 1.28.3

